Kwonyoung Ryu

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Education

2021 – ongoing **Ph.D. Pohang University of Science and Technology** in Graduate School of Artificial Intelligence.
Research Interest : Multi-view Image 3D Segmentation, Generalized 3D (LiDAR/Image) perception, Autonomous driving perception, Occupancy Prediction, Feed-forward Gaussian Splatting, Amodal Scene reconstruction
Supervisor : Sunghyun Cho, Co-advisor : Jaesik Park
*Currently working at **Seoul National University** (2023.09 – ongoing, Advisor : Jaesik Park)

2019 – 2021 **M.S. Korea Advanced Institute of Science and Technology** in Mechanical Engineering.
Thesis title: *Scanline Resolution-Invariant Depth Completion Using a Single Image and Sparse LiDAR Point Cloud* (Supervisor : Kuk-jin Yoon).

2014 – 2019 **B.S. Korea Advanced Institute of Science and Technology** in Mechanical Engineering.

Research Publications

Conference Proceedings

1. I.-J. Lee, N. Kim, R. Exley, K. Ryu, and J. Park, "AmodalBox: Automated 3D bounding box annotation under partial visibility," in *Conf. on Robot Learning (CoRL)*, 2026.
2. K. Ryu, I.-J. Lee, J. Jin, H. Lee, J. Lee, and J. Park, "Extending a large view synthesis model for multi-view panoptic segmentation," in *European Conf. on Computer Vision (ECCV)*, 2026.
3. I.-J. Lee, M. Kim, K. Ryu, P. Musacchio, and J. Park, "Openbox: Annotate any bounding boxes in 3d," in *Int. Conf. on Neural Information Processing Systems (NeurIPS), Highlight*, 2025.
4. K. Ryu*, S. Hwang*, and J. Park, "Instant domain augmentation for lidar semantic segmentation," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 9350–9360.

Journal Articles

1. K. Ryu, K.-i. Lee, J. Cho, and K.-J. Yoon, "Scanline resolution-invariant depth completion using a single image and sparse lidar point cloud," *IEEE Robotics and Automation Letters*, vol. 6, no. 4, pp. 6961–6968, 2021.
2. T. Kim, K. Ryu, K. Song, and K.-J. Yoon, "Loop-net: Joint unsupervised disparity and optical flow estimation of stereo videos with spatiotemporal loop consistency," *IEEE Robotics and Automation Letters*, vol. 5, pp. 5597–5604, 2020.

ArXiv

1. K. Lee, I.-J. Lee, M. Kwak, K. Ryu, J. Hong, and J. Park, *SpatialMosaic: A Multiview VLM Dataset for Partial Visibility*. 2025, vol. abs/2512.23365. arXiv: 2512.23365. URL: https://arxiv.org/abs/2512.23365.

2 W. Dong*, K. Ryu*, M. Kaess, and J. Park, *Real-time Registration and Reconstruction with Cylindrical LiDAR Images*. 2021, vol. abs/2112.02779. arXiv: 2112.02779. URL: https://arxiv.org/abs/2112.02779.

Miscellaneous Experience

Academic Services

Conference Reviewer — CVPR(23-26), ECCV(22,24,26), ICRA(22-23), IROS(24-25), NeurIPS(25-26), BMVC(26)

Reviewer Award — Top Reviewer Awards, NeurIPS 2025

Talk

2023.11 — Towards Generalizable Perception for LiDAR / outdoor 3D pointcloud, LG AI Seminar

Teaching

2022.09 — **Teaching Assistant**. Topics in AI:3D Vision, Prof. Jaesik Park

2023.03 — **Head Teaching Assistant**. Deep Learning, Prof. Namhoon Lee